

# RETAIL DEMAND INTELLIGENCE: FORECASTING & DEMAND DRIVER ANALYSIS

Inventory Planning, Pricing Dynamics  
& Operational Performance

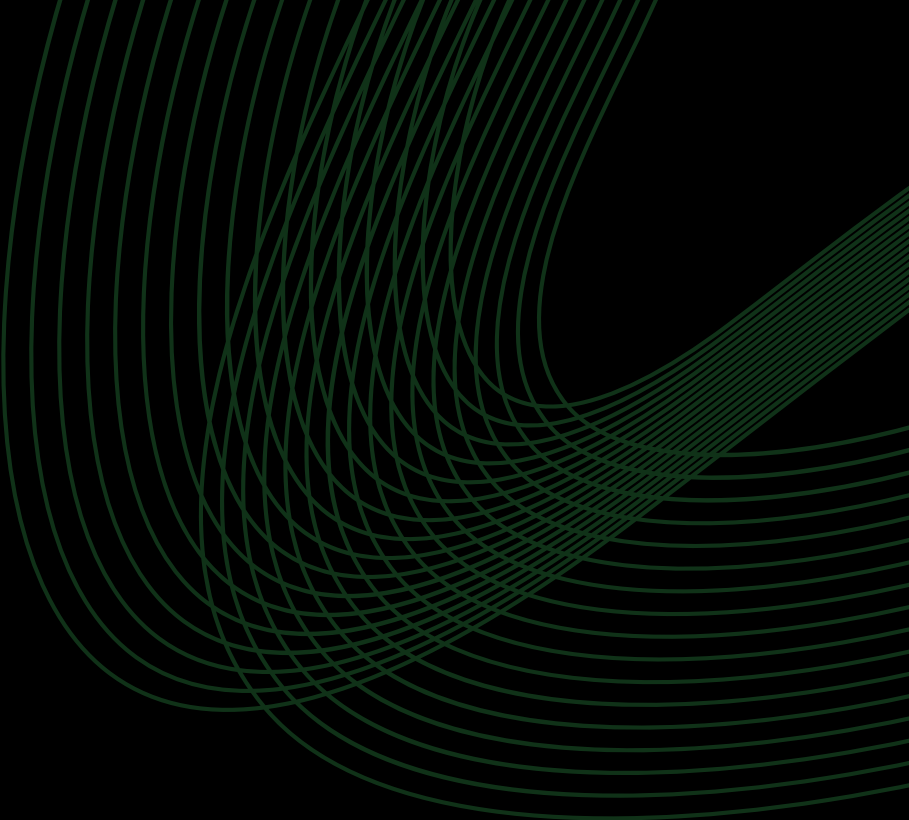


## TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &  
ML libraries

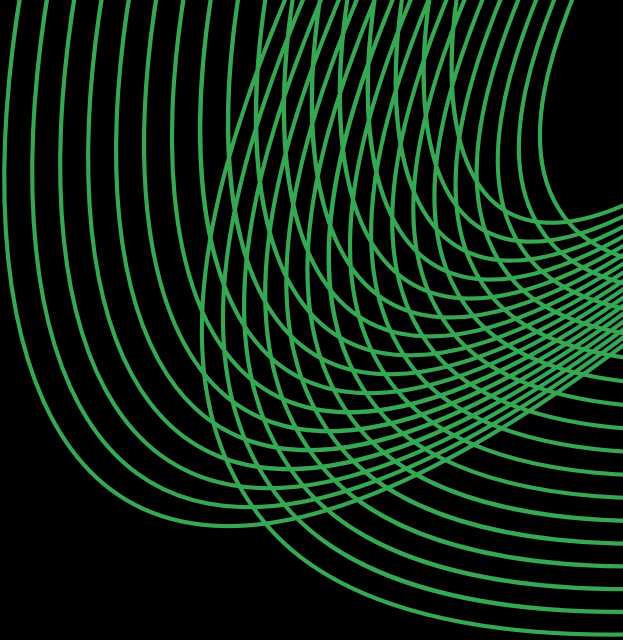
## PRESENTED BY

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# Overview & Business Objective



# 1.1 PROJECT OVERVIEW

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

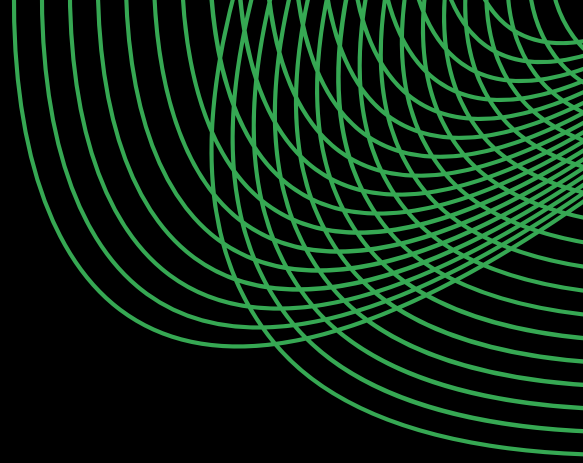
The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.



# 1.2 PROBLEM STATEMENT



Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

***How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?***

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.



# 1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.



## 1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, demand modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

### Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

### Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

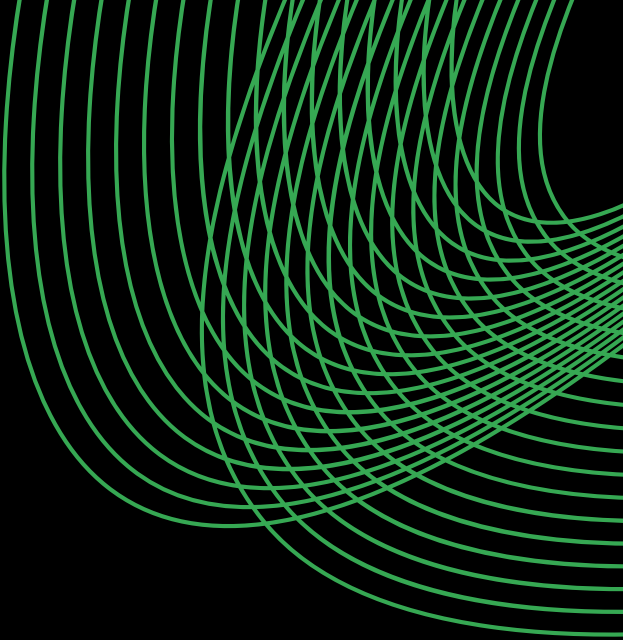
### Explainability

- Feature importance analysis
- Original-feature aggregation
- Investigation of large prediction errors

04

### Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning



# Data Understanding & Exploratory Data Analysis





## 2.1 DATA OVERVIEW

The dataset provides a comprehensive view of retail operations across stores and products, covering inventory levels, sales activity, ordering patterns, pricing, discounts, promotions, weather conditions, seasonality, and demand. These variables provide the foundation for analyzing demand behavior, identifying operational patterns, and developing predictive models for retail demand.

Data Source: Kaggle

### Dataset Size

76,000

Records

16

Original Features

### Purchase Diversity

4

Regions

5

Stores

20

Products



# DATA OVERVIEW

Below is a detailed description of the feature set:

| Dataset Features   | Type                   | Feature Description                                         |
|--------------------|------------------------|-------------------------------------------------------------|
| Date               | Date / Time            | Date of the observation                                     |
| Store ID           | Categorical            | Unique identifier of the store                              |
| Product ID         | Categorical            | Unique identifier of the product                            |
| Category           | Categorical            | Product category                                            |
| Region             | Categorical            | Geographic region of the store                              |
| Inventory Level    | Numerical (Discrete)   | Inventory available at the time of observation              |
| Units Sold         | Numerical (Discrete)   | Number of units sold                                        |
| Units Ordered      | Numerical (Discrete)   | Number of units ordered                                     |
| Price              | Numerical (Continuous) | Selling price of the product                                |
| Discount           | Numerical (Discrete)   | Discount applied to the product                             |
| Weather Condition  | Categorical            | Weather condition during the observation                    |
| Promotion          | Binary                 | Indicates whether a promotion was active                    |
| Competitor Pricing | Numerical (Continuous) | Price of the comparable product offered by competitors      |
| Seasonality        | Categorical            | Seasonal classification of the observation                  |
| Epidemic           | Binary                 | Indicates whether an epidemic-related condition was present |
| Demand             | Numerical (Discrete)   | Target variable representing product demand                 |

## 2.2 TARGET VARIABLE

The target variable selected for prediction is:

*Demand*

Demand represents the quantity that the model attempts to estimate based on the available information

The target has:

**104.32**

Mean

**100**

Median

**4**

Minimum

**430**

Maximum

**46.96**

Standard Deviation

The distribution is concentrated around approximately 70-130 units, with a noticeable right tail extending toward higher demand values.



## 2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The *Date* variable was converted into usable temporal components, while categorical variables were prepared for machine learning using one-hot encoding.

